Unit 4

Renewable Power Generation

What is renewable energy resources?

Renewable energy is energy derived from natural sources that are replenished at a higher rate than they are consumed.

Eg: Sunlight and wind are such sources that are being replenished.

Advantages of Renewable energy sources:

- Clean, environment friendly
- Available in ample amount, non depleting
- Needs less maintenance
- Creates Job opportunities
- Locally available, satisfy rural energy needs
- Increase energy Security
- Stable energy Prices makes it more favorable to use
- Do not emit green house gas

Disadvantages of Renewable energy sources:

- Unreliable
- Lower efficiency
- Need large capital money
- Hamper electrical system stability.

What is a PV solar power plant?

Photovoltaic power plants use large areas of photovoltaic cells, known as PV or solar cells, to convert sunlight into usable electricity. These cells are usually made from silicon alloy.

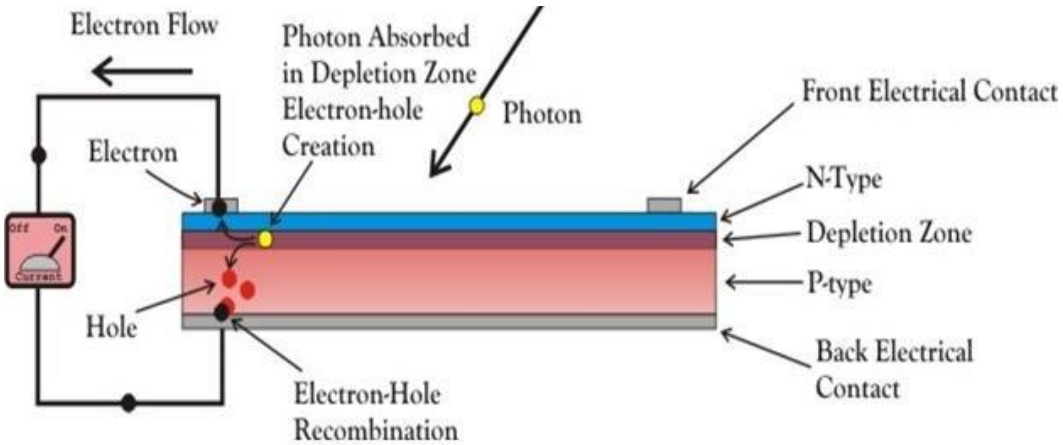
How does a solar PV power plant work?

- Solar PV power plants work in the same manner as smaller domestic-scale PV panels.

- As we have seen, most solar PV panels are made from semiconductor materials, usually some form of silicon. When photons from sunlight hit the semiconductor material, free electrons are generated which can then flow through the material to produce a direct electrical current.
- This is known as the photoelectric effect. The DC current then needs to be converted to alternating current (AC) using an inverter before it can be directly used or fed into the electrical grid.
- PV panels are distinct from other solar power plants as they use the photo-effect directly, without the need for other processes or devices.

Photovoltaic plants

- A photovoltaic cell, commonly called a solar cell or PV, is a technology used to convert solar energy directly into electricity. A photovoltaic cell is usually made from silicon alloys.
- Particles of solar energy, known as photons, strike the surface of a photovoltaic cell between two semiconductors.
- These semiconductors exhibit a property known as the photoelectric effect, which causesthem to absorb the photons and release electrons. The electrons are captured in the formof an electric current - in other words, electricity.



Working of Solar Power Plant

- As sunlight falls over a solar cells, a large number of photons strike the p-type region ofsilicon. Electron and hole pair will get separated after absorbing the energy of photon.The electron travels from p-type region to n-type region due to the action of electric field at p-n junction. Further the diode is reversed biased to increase this electric field.So this current starts flowing in the circuit for individual solar cell. We combine the current of all the solar cells of a solar panel, to get a significant output.
- Solar power plant have a large number of solar panels connected to each other to get alarge voltage output. The electrical energy coming from the combined

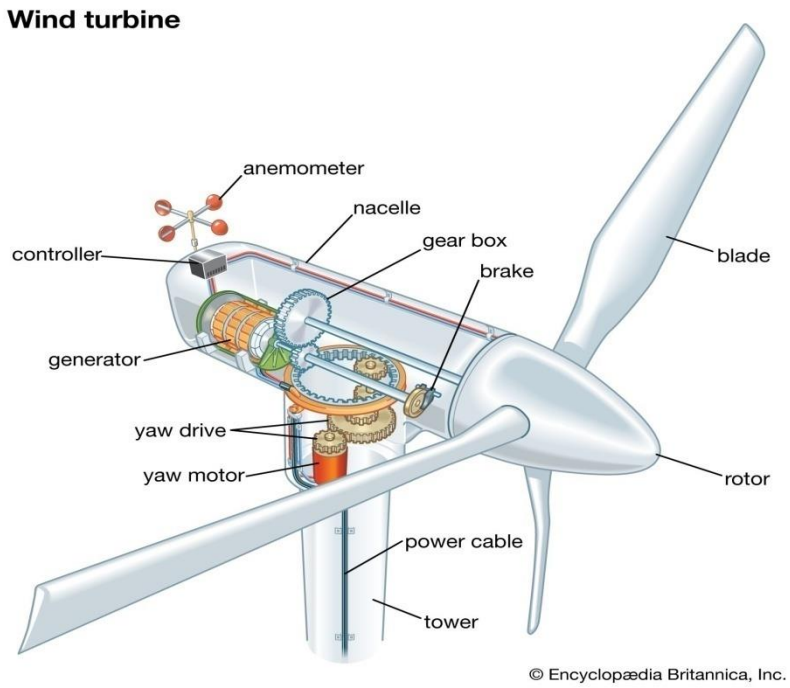
effort of solar panels is stored in the Lithium ion batteries to be supplied at night time, when there is no sunlight.

Solar Collector

Solar Collectors are of two types:

- Flat Plate collector
- Concentrating Type Solar Collector

2. WIND ENERGY



Main components

1.Rotar

The rotor **is** the rotating part of the turbine; it consists of three blades and a central part connecting the blades**,** the hub**.** Although it is the most common, **a** turbine does not necessarily have three blades. The blades aren't strong; they're hollow and made of composite material that's both lightweight and strong. The trend is to make them bigger (for more power), lighter, and more robust. For aerodynamics, the blades are shaped like an airfoil (like an airplane's wings).

Also, they are not flat and have a twist between their root and tip. The blade can rotate up to 90° around its axis. This movement is called pitch.

2. **Anemometer:**
Measures the wind speed and spreads wind speed data to the organizer.

3. Blades:
Lifts and rotates subsequently when wind is blown over them, causing the rotor to spin. Maximum turbines have either 3 to 2 blades.

4. Nacelle:
Sits at top the tower and contains the gear box, low- and high-velocity shafts, generator, regulator, and brake.

5.Tower
Wind turbine tower are formed at various different altitudes, with the average being unevenly 50m and the highest surpassing 200m.
Tower is equipped with service platform and additional platforms. Each tower section is illuminated by means of ladders.
They are frequently coated with a zinc-based appearance and epoxy and urethane coatings to provide electromagnetic resistance.

6. Gear box
Gear connect the low-speed shaft to the high speed shaft and increase rotational speeds from about 30 to 60 rotations per minute (rpm) to about 1000 to 1800 rpm, the rotational speed required by most generators to produce electricity.

7. Brake
A disc brake, which can be applied mechanically, electrically, or hydraulically to stop the rotor in emergencies.

8.High speed shaft

Drives the generator.

9.Low-speed shaft

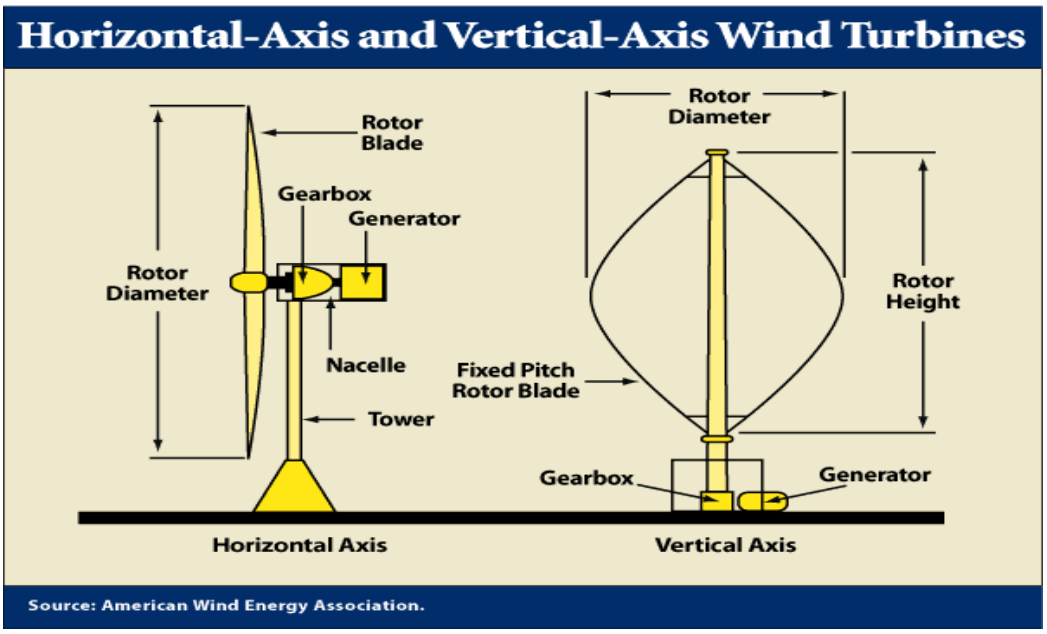
The rotor turns the low-speed shaft 30 to 60 rotations per minute.

10. Yaw Drive

Upwind the turbine face into the wind, the yaw drive is used to keep the rotar facing into the wind direction changes.

Types Of Wind Turbines

1. Horizontal axis type
2. Vertical axis type

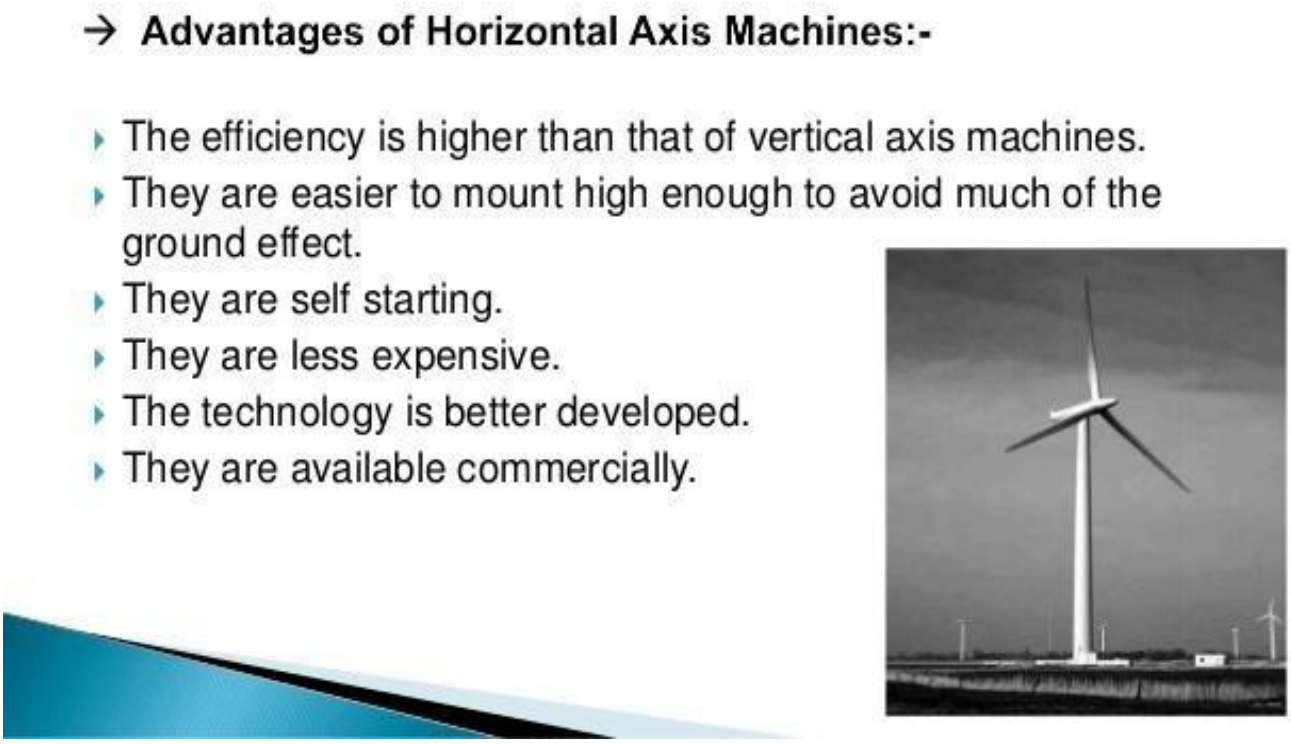


1. **Horizontal Axis Turbine**

Horizontal axis wind turbines are the most commonly used type of wind turbine. They can be defined as the turbine in which the shaft of the rotor is in the direction of the wind. The wind turbine could be single-bladed, double-bladed, or three-bladed. HAWTs are the most common wind machine designs in use today.

Horizontal axis wind turbine consists of nacelle as main equipment which is mounted on a top of a vertical tower. Nacelle contains a hub in front in which blades are mounted. Orientation of the blades is along horizontal axis that's why called as HAWT. As the wind flows the blades starts rotating .These blades are connected with main shaft. The rotation of main shaft is equal to the rotation of blades which is less in no. and can produce less power. For increasing the number of rotation main shaft is connectedto a gear box in which the number of

rotations increases by 100 times. These gears are connected with another shaft which rotates with 100 times greater speed with respect to main shaft. This shaft is also called high speed shaft which is connected to aero turbine which produces mechanical power.



Disadvantages of HAWT

- Heavy (more parts than VAWT)
- Difficult to install and transport
- High installation and transportation cost

3. Vertical Axis Wind Turbine

Vertical Axis Wind Turbine, or VAWT, is probably the oldest type of windmills in which the axis of the drive shaft is perpendicular to the ground. It is a type of windmill where the main rotor shaft runs vertically, as opposed to the horizontal axis wind turbine. The blades of the VAWTs rotate with respect to their vertical axes that are perpendicular to the ground. They can capture wind from any

direction and their heavy machinery is at ground level. Since the machinery is set on the ground, it simplifies the wind tower design and construction and reduces the turbine cost consequently. Unlike HAWTs, VAWTs are typically used in areas with turbulent wind flow such as coastlines, rooftops, cityscapes, etc.

Advantages of VWAT

- These turbines have fewer parts than those that orient the rotary mechanism and blades horizontally. That means fewer components to wear out and break down.
- The supporting strength of the tower doesn't need to be as much, because the gearbox and generator are near the ground. Parts for controlling pitch and yaw aren't needed either.
- The turbine doesn't have to be facing the right wind direction either. In a vertical system, air flowing from any direction or speed can rotate the blades. Therefore, the system can be used to generate power in gusty winds and when they're blowing steadily.
- Maintenance workers do not have to climb as high to reach parts of the tower.
- Cheaper to produce than horizontal axis turbines.
- More easily installed compared to other wind turbine types.

Disadvantages of VWAT

- Since the wind turbines are lower to the ground, they do not harness the higher wind speeds often found at higher levels.
- Vibration can be an issue at times, and even increase the noise produced by the turbine.
- The overall efficiency is much lower than horizontal axis wind turbine.

Horizontal
VS
Vertical Axis Wind Turbine

Comparison Chart

Horizontal Axis Wind Turbine	Vertical Axis Wind Turbine
The rotating axis of the blades is parallel to the direction of the wind.	The rotating axis of the blades is perpendicular to the direction of the wind.
The main rotor shaft runs horizontally in HAWTs.	The main rotor shaft runs vertically in VAWTs.
HAWTs are generally used under streamline wind conditions where a constant stream and direction of wind is available.	VAWTs are mainly beneficial in areas with turbulent wind flow such as rooftops, coastlines, cityscapes, etc.
The rotor faces the wind stream to capture maximum wind energy.	The rotor can accept wind stream from any direction.
Inspection and maintenance is difficult in HAWT.	Inspection and maintenance is easy.
HAWTs extract more power from wind.	VAWTs extract less power from wind.
They are more efficient than VAWTs.	They are less efficient than HAWTs.
They operate fine in moderate wind speeds.	They can operate even in low wind speeds.

DifferenceBetween.net

What is Biogas?

Biogas refers to a mixture of gases produced by the anaerobic decomposition of organic matter such as agricultural waste, municipal waste, plant residue, food waste etc. Biogas consists of methane, carbon dioxide, a small amount of hydrogen sulphide, and moisture.

What is a Biogas Plant?

The biogas plant consists of a dome-like structure. Organic materials such as discarded food residue, fats, sludge, cow dung etc. are mixed with water and fed to the digester through the inlet as shown in the figure.

The digester is a sealed chamber where the anaerobic decomposition of organic matter takes place. After a few days, the organic matter completely decomposes to generate gases like methane, carbon dioxide, hydrogen and hydrogen sulphide. These gases are drawn through pipes from the storage tank above the digester and distributed through decentralization channels to nearby centres for use.

Advantages of Biogas

- **Non-polluting:**

  Biogas burns without smoke; hence no harmful gas such as CO_2, CO, NO_2, and SO_2 are evolved.

- **Reduces Landfills:**

  The slurry produced after the production of biogas is used as manure in fields. The method of disposal is safe and efficient and hence no space is wasted in the form of landfills.

- **Cheaper technology:**

  Biogas plants require very little installation cost and become self-sufficient in a span of 3-4 months.

- **Generates employment:**

Work opportunity for thousands of people is created, especially in rural areas.

- **Renewable source of energy:**

  It is considered as a renewable source of energy because the production is dependent on the generation of waste which is an endless process.

Disadvantages of Biogas

- **Not efficient enough on a large scale:**

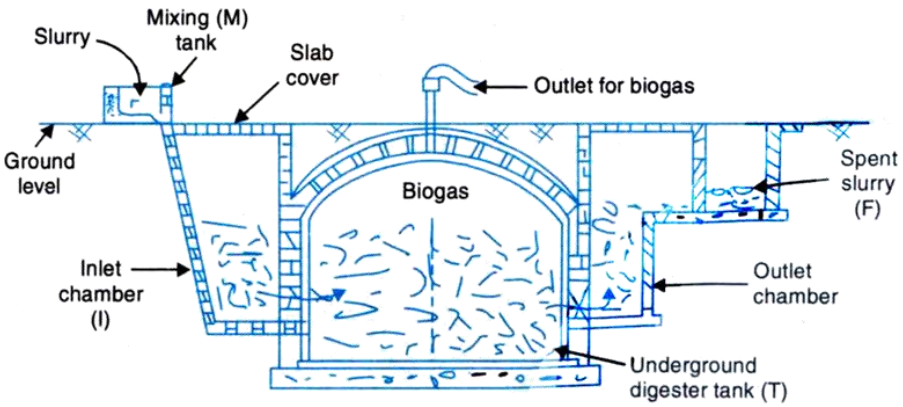
  Since it is difficult to enhance the efficiency of biogas, it is not economically viable to use biogas on a large scale.

- **Contains impurities:**

  It contains many impurities that are difficult to control even after purification rounds. Biogas when compressed, to be used as fuel, proves to be highly corrosive to the container

- **Unstable and hazardous:**

  When methane comes in contact with oxygen, it reacts violently to produce carbon dioxide. The highly inflammable nature of methane makes it prone to explosions.



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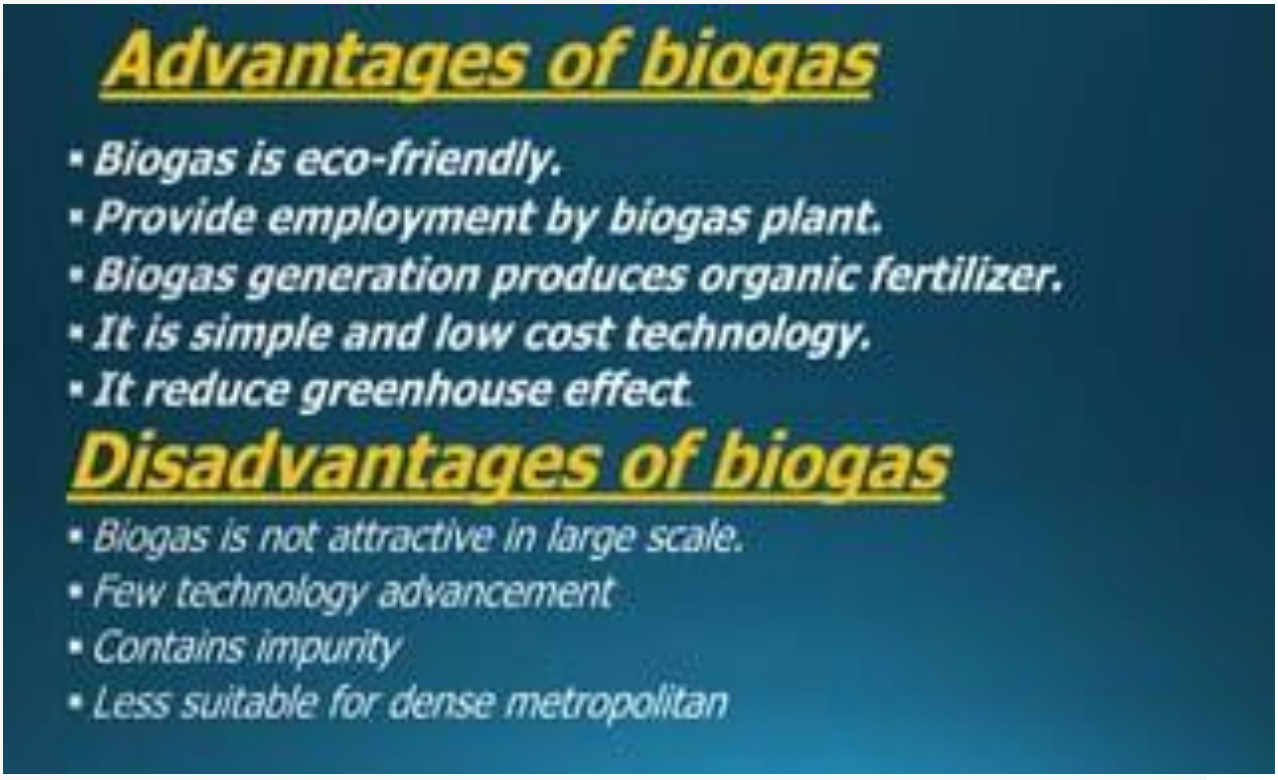
Working of Fixed Dome type Biogas Plant

- The various forms of biomass are mixed with an equal quantity of water in the mixing tank. This forms the **slurry**.
- The slurry is fed into the digester through the inlet chamber.
- When the digester is partially filled with the slurry, the introduction of slurry is stopped and the plant is left unused for about two months.
- During these two months, anaerobic bacteria present in the slurry decomposes or ferments the biomass in the presence of water.
- As a result of anaerobic fermentation, biogas is formed, which starts collecting in the dome of the digester.

- As more and more biogas starts collecting, the pressure exerted by the biogas forces the spent slurry into the outlet chamber.
- From the outlet chamber, the spent slurry overflows into the overflow tank.
- The spent slurry is manually removed from the overflow tank and used as manure for plants.
- The gas valve connected to a system of pipelines is opened when a supply of biogas is required.
- To obtain a continuous supply of biogas, a functioning plant can be fed continuously with the prepared slurry.

Site Selection For Biogas Plant

- Land surface should be flat where we want to install the biogas plant.
- Soil of land should not be loose.
- Biogas should be near to home or farm.
- Biogas plant should be near to cattle shade so that cow dung receives easily.



BIOMASS

- Biomass is organic matter from plants, micro--organism grown on land and water and their derivatives. The energy obtain from biomass is also called the biomass energy. It is a renew--able energy source. Because organic matter generated everyday. Coal, petroleum, natural gas are not come in biomass category because they produce from dead, burried biomass under high pressure and temperature during several millions of year.

Ocean Thermal Energy (OTEC)

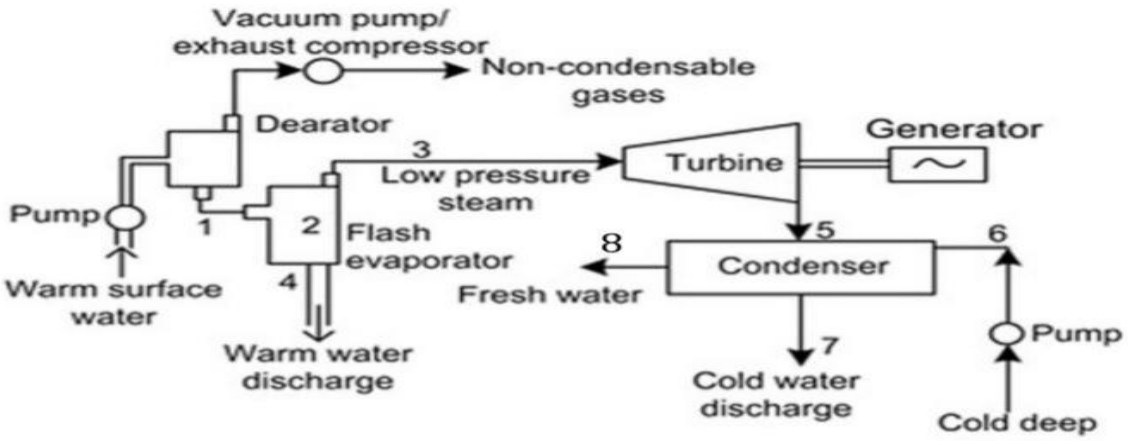
- Ocean thermal energy conversion is an electricity generation system.
- Ocean Thermal Energy, also called Ocean Thermal Energy Conversion (OTEC), refers to using the temperature difference between the deep parts of the sea, which are cold and the shallow parts of the sea, which are cold, to run a heat engine and produce useful work.
- The deeper parts of the ocean are cooler because the heat of sunlight cannot penetrate very deep into the water.
- Here the efficiency of the system depends on the temperature difference.
- Greater the temperature difference, the greater the efficiency.
- The temperature difference in the oceans between the deep and shallow parts is maximum in the tropics, 20° C to 25° C.
- Tropics receive a lot of sunlight which warms the surface of the oceans, increasing the temperature gradient.

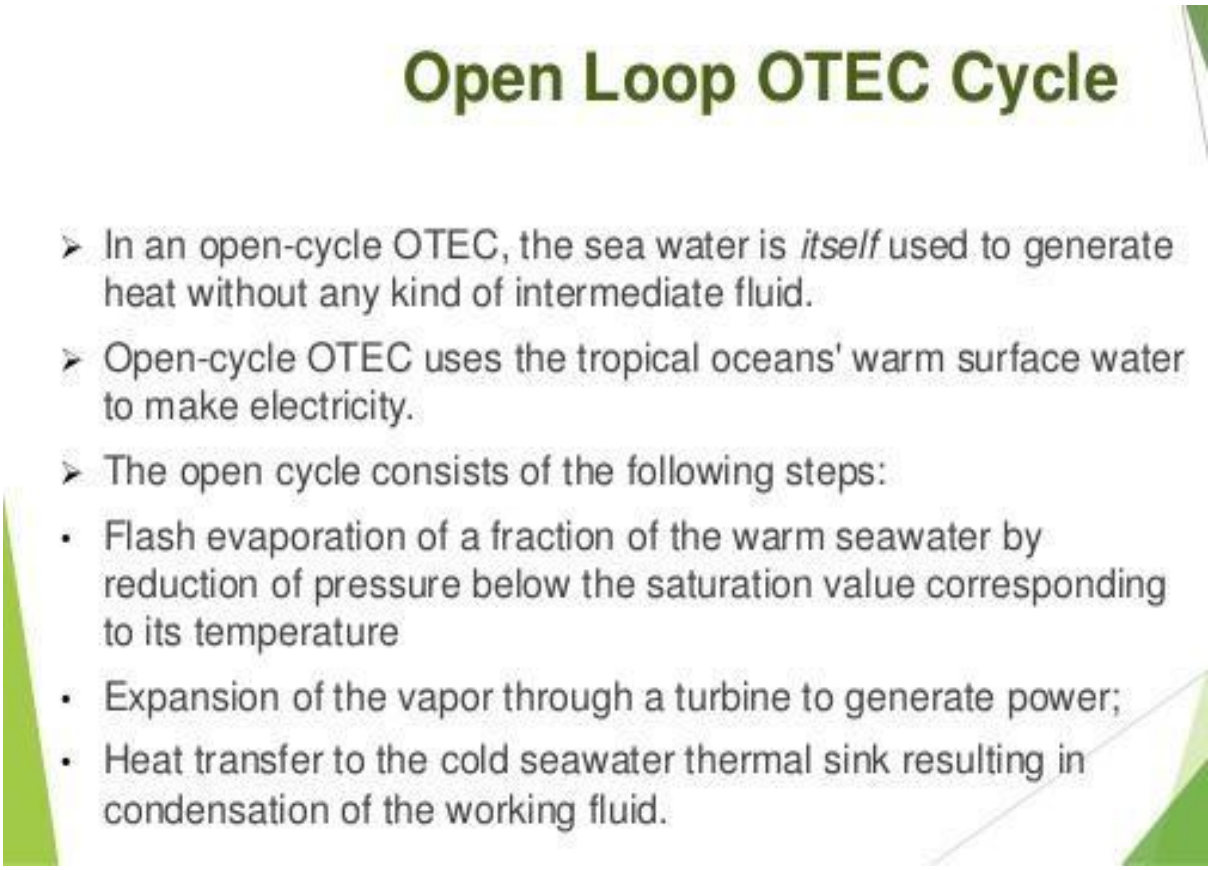
Types of OTEC Power Plant:

There are basically three types of OTEC systems developed that can utilize sea water temperature differentials.

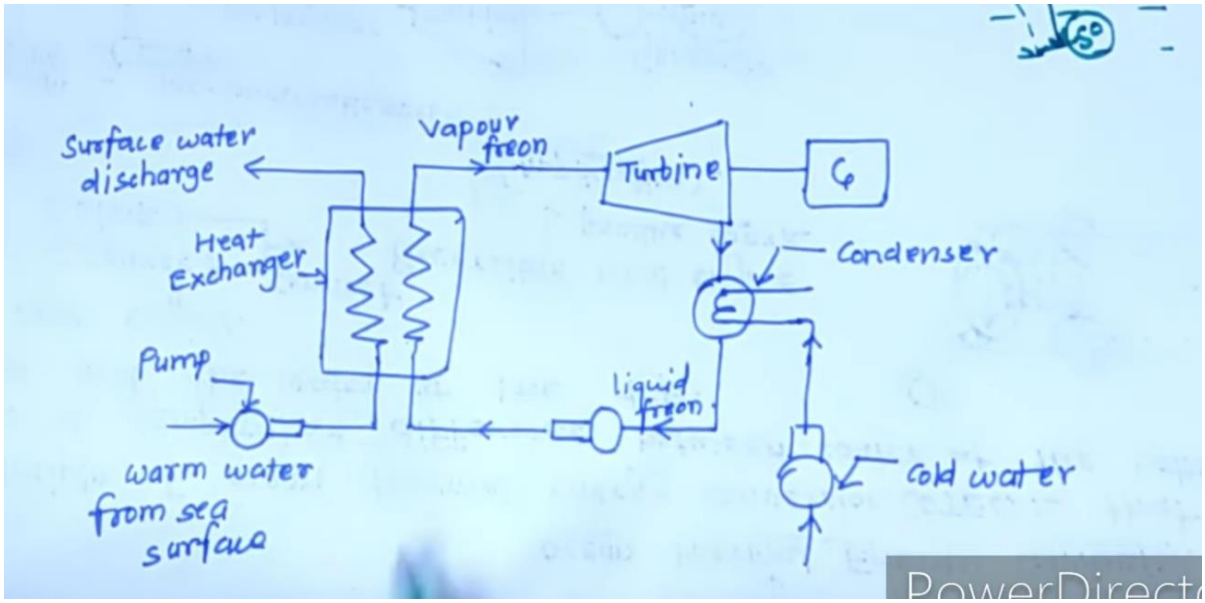
- Claude cycle or Open cycle OTEC
- Andersons cycle or Closed cycle OTEC
- Hybrid cycle

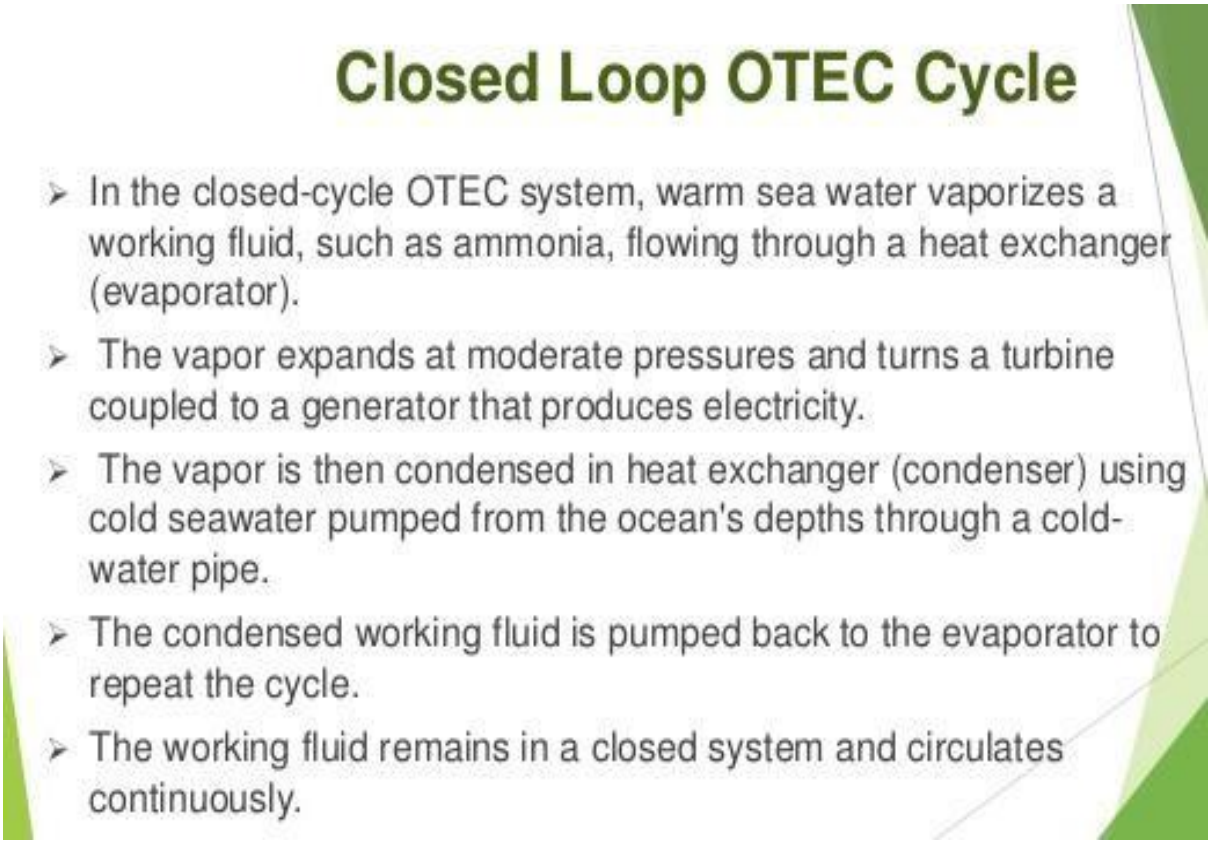
1. **Open cycle OTEC**





2. **Closed cycle OTEC**





Advantages of OTEC

The main advantages of OTEC system are;

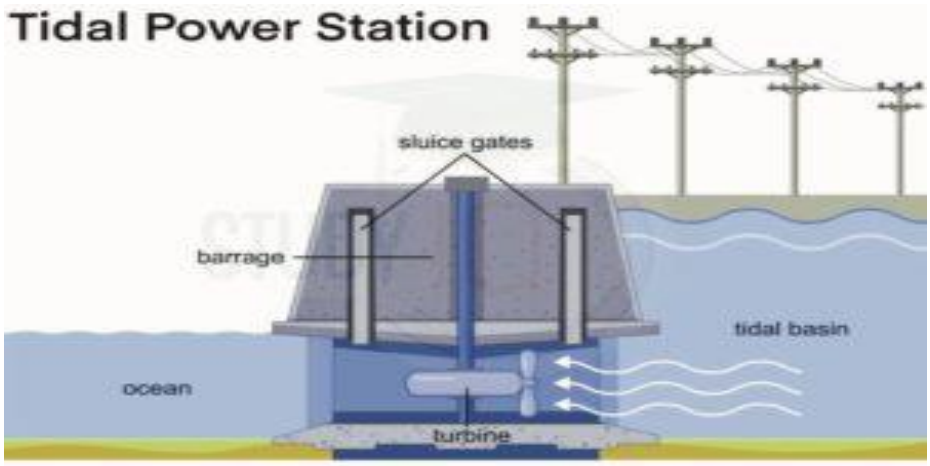
- Renewable source

- No emissions or air pollution,
- Consistent energy supply,
- Low complexity
- Low environmental impact

Disadvantage of OTEC

- Large water volume.
- Low thermal energy density.
- Marine pollution risk.
- Geographic limitation and developmental issues.
- It is not fully developed technology.

TIDAL POWER PLANT



Components of tidal power plant

1. **Dam or barrage**: A dam is used to form a barrier between the sea and a basin. It is designed to withstand the force of tidal waves and typically consists of reinforced concrete with channels for turbines**.**

2. **Sluice ways** are used to fill the basin during the high tide or empty the basin during the low tide, as per operational requirement. These devices are controlled through gates.

3. **Tidal generator:** Tidal stream generators are very similar to wind turbines except their below the water surface instead of above or on land.

   The turbine and generator converts the movement of water coming from change in tide , the kinetic energy, into electricity.

   Water is 830 times denser than air and therefore can generate electricity at lower speeds than wind turbines.

Advantages of Tidal Energy

- It is an inexhaustible source of energy.
- Tidal energy is environment friendly energy and doesn't produce greenhouse gases.
- As 71% of Earth's surface is covered by water, there is scope to generate this energy on large scale.
- Efficiency of tidal power is far greater as compared to coal, solar or wind energy. Its efficiency is around 80%.
- Tidal Energy doesn't require any kind of fuel to run.
- The life of tidal energy power plant is very long.
- The large density of water, almost 1000 times greater than in air, results in very large amounts of energy to get out of the tidal currents even if the speed is low.

Disadvantages of Tidal Energy

- Cost of construction of tidal power plant is high.
- There are very few ideal locations for construction of plant and they too are localized to coastal regions only.
- Intensity of sea waves is unpredictable and there can be damage to power generation units.
- Influences aquatic life adversely and can disrupt migration of fish.
- The actual generation is for a short period of time. The tides only happen twice a day so electricity can be produced only for that time.
- Usually the places where tidal energy is produced are far away from the places where it is consumed. This transmission is expensive and difficult.